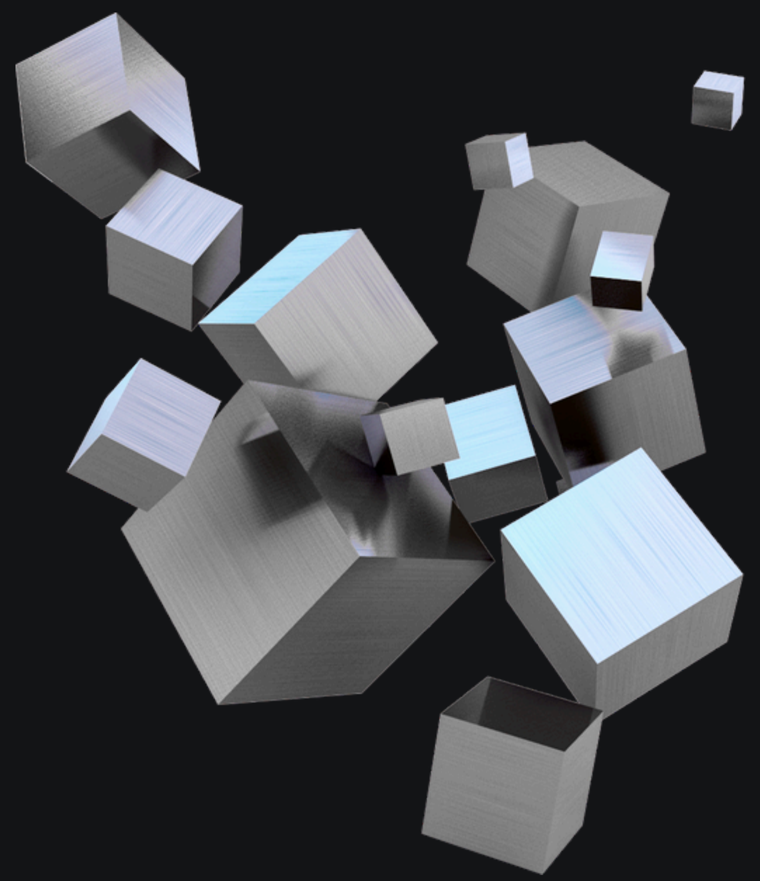
Career path Recommender System



Presented by:

Rana Alanezi - Ghaida Alharbi
Lama Alhunayhin - Noura Alyemni

Group 7
Course: SWE485

Problem & Business Value

The Problem

- Students feel lost after graduation.
- Job descriptions are unclear.
- No system links skills → real careers.

Our Solution

An intelligent advisor that analyzes:
Student skills
Market jobs
Hidden patterns
And suggests the best career path

Business Value

More accurate decisions
Less trial and error
Scalable for universities

Data Overview

1

Data Source

[Job Skill Set Dataset (Kaggle)]

2

Dataset Size & Structure

~ 1,100 job postings

5 columns →

- Job_id
- category
- job_title
- job_description
- job_skill_set

3

Key Features

Job descriptions, required skills

4

Collection Method

Collected from real online job postings

Process / Approach

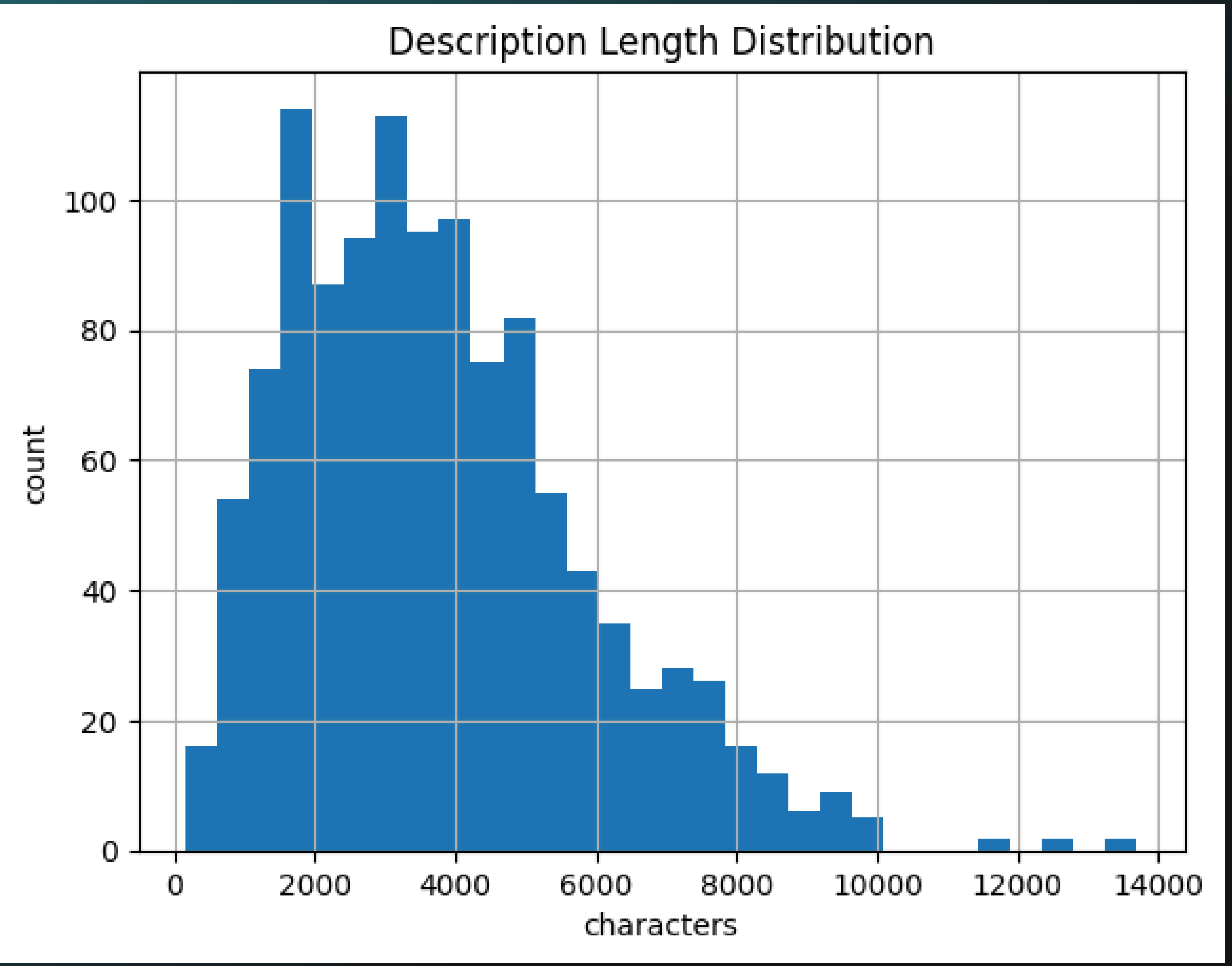
1. Data Cleaning → 2. Exploratory Data Analysis (EDA) → 3. Supervised Classification → 4. Unsupervised Clustering → 5. Generative AI Integration

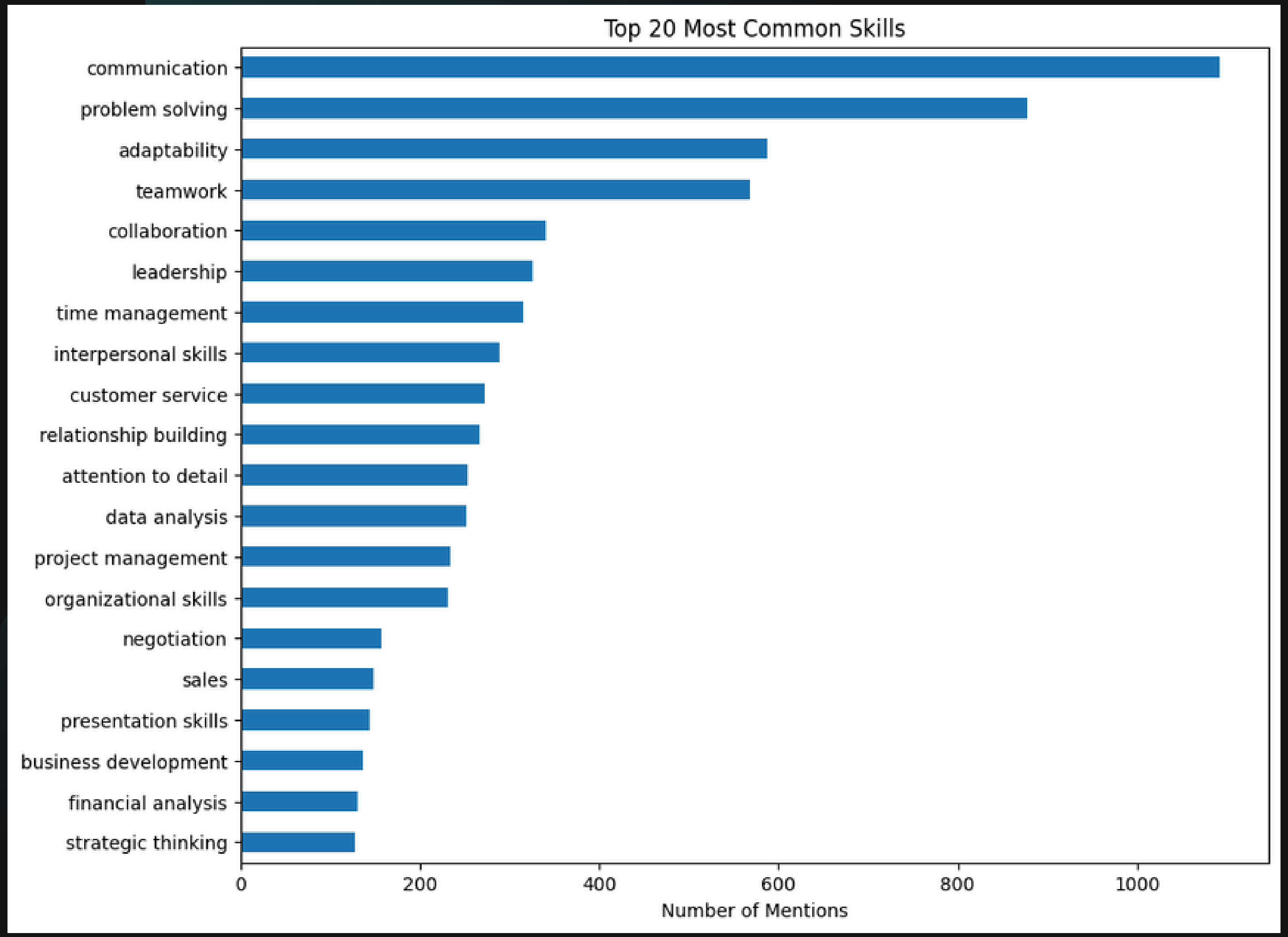
Tools Used

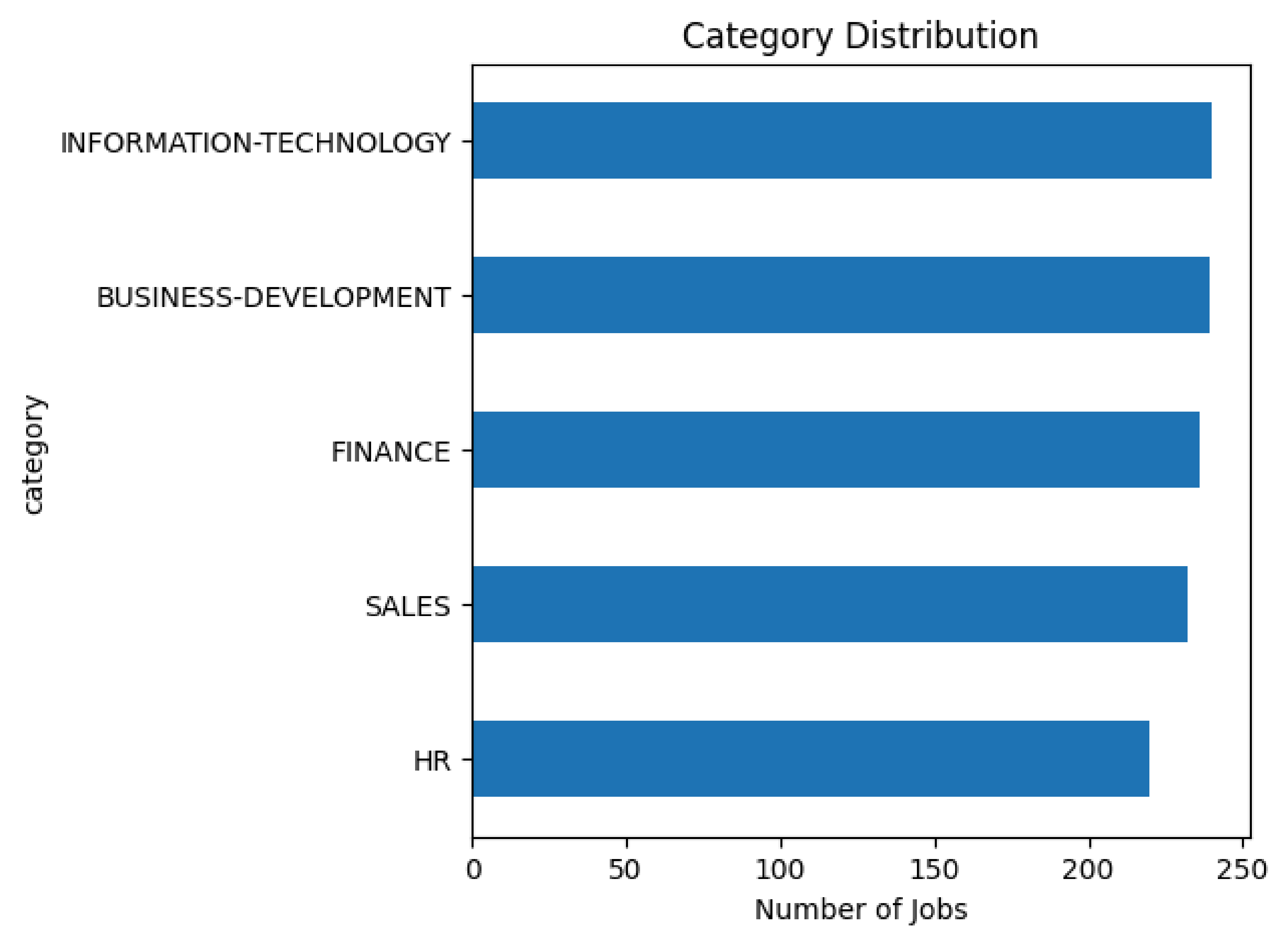
1. Python
2. Pandas
3. Scikit-learn
4. OpenAI API
5. Jupyter Notebook

Phase 1 (Insights): Brief Look at Phase 1 Analysis

In Phase 1, we started by exploring the dataset to understand its structure before training any model.
First, we looked at the class balance to check whether the categories were evenly distributed or if one class dominated the dataset.
This is important because an imbalanced dataset can cause the model to become biased.
Next, we identified the top skills, which represent the most frequent keywords appearing in the data.
These skills help us understand what features are most relevant for distinguishing between the classes.







Class Distribution : The dataset is well-balanced across all five categories. The distribution is nearly even, with the highest class (IT at 240) and the lowest (HR at 220) being very close. This is ideal for the modeling phase as it prevents model bias.

Phase 2 (Model Comparison): Naive Bayes vs Linear SVM classification report

In Phase 2, we compared Naive Bayes and Linear SVM.
The results showed that Linear SVM performed better, offering higher accuracy and more stable predictions.
The F1-Score confirms this:
SVM scored 0.9880, while Naive Bayes scored 0.9839.
Even though the difference is small, SVM handled the high-dimensional text features more effectively.
In short, SVM was the stronger and more reliable model."

```
Accuracy - Naive Bayes: 0.8718
Accuracy - SVM: 0.9530
```

Classification Report - Naive Bayes

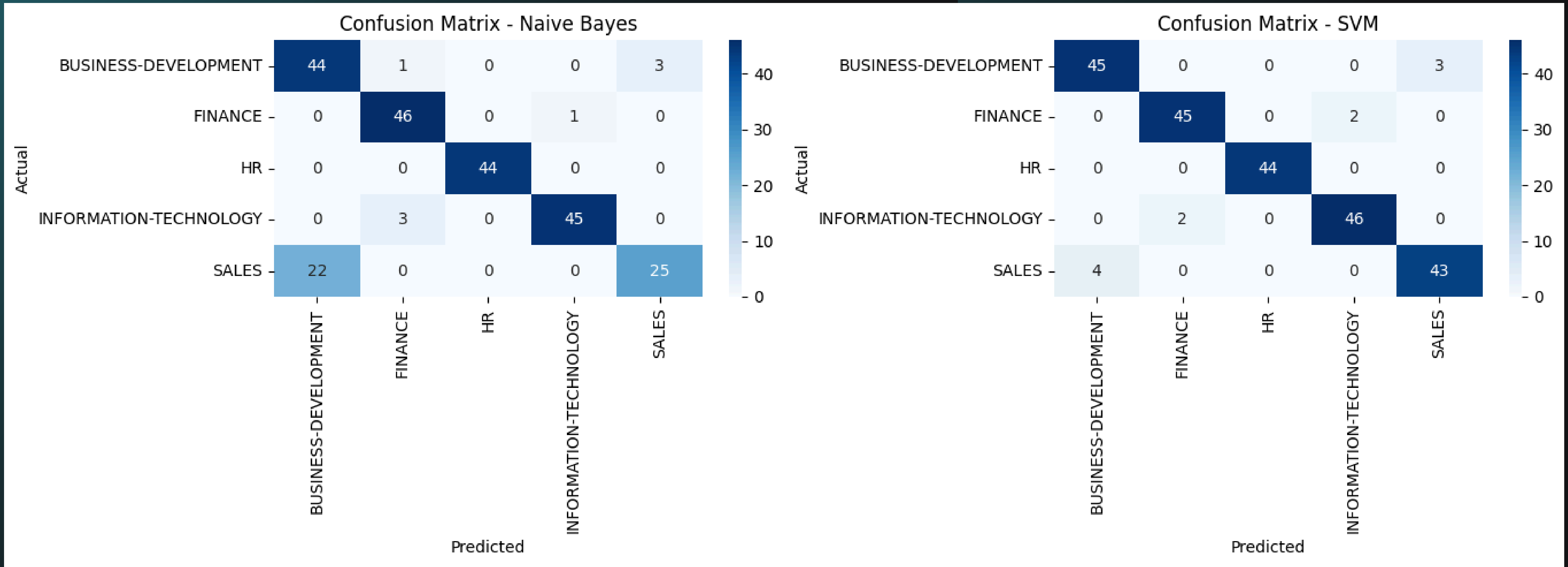
	precision	recall	f1-score	support
BUSINESS-DEVELOPMENT	0.67	0.92	0.77	48
FINANCE	0.92	0.98	0.95	47
HR	1.00	1.00	1.00	44
INFORMATION-TECHNOLOGY	0.98	0.94	0.96	48
SALES	0.89	0.53	0.67	47
accuracy			0.87	234
macro avg	0.89	0.87	0.87	234
weighted avg	0.89	0.87	0.87	234

Classification Report - SVM

	precision	recall	f1-score	support
BUSINESS-DEVELOPMENT	0.92	0.94	0.93	48
FINANCE	0.96	0.96	0.96	47
HR	1.00	1.00	1.00	44
INFORMATION-TECHNOLOGY	0.96	0.96	0.96	48
SALES	0.93	0.91	0.92	47
accuracy			0.95	234
macro avg	0.95	0.95	0.95	234
weighted avg	0.95	0.95	0.95	234

Phase 2 (Confusion Matrix Deep Dive: Naive Bayes vs Linear SVM

To understand why SVM outperformed Naive Bayes, we analyzed the confusion matrix.
SVM was able to clearly separate the ‘Sales’ and ‘Business Development’ categories, even though they are very similar.
On the other hand, Naive Bayes struggled and frequently confused these two classes.
This happens because Naive Bayes relies on simple probability assumptions, while SVM can capture more complex decision boundaries.
In summary, SVM succeeded because it learned the patterns that truly differentiate the classes, while Naive Bayes did not."



Phase 3: Discovering Hidden IT Specializations

For Phase 3, the focus is on enhancing the model by discovering hidden sub-groups **inside a single category (Information-Technology).**

Full Dataset → Filter IT Jobs → Remove Job Titles → TF-IDF (Term Frequency- Inverse Document Frequency) Vectorization → K-Means (k=5)

Cluster 0: Security

Top Keywords

['security', 'network', 'information', 'management', 'information security', 'cisco', 'network security', 'infrastructure', 'systems', 'firewalls']

Cluster 1: Project Management

Top Keywords

['project', 'management', 'data', 'business', 'experience', 'project management', 'technology', 'skills', 'financial', 'development']

Cluster 2: Internships

Top Keywords

['year', 'placement', 'program', 'internship', 'professional', 'interviewing', 'educational', 'networking', 'college', 'students']

Cluster 3: IT Support

Top Keywords

['support', 'software', 'hardware', 'troubleshooting', 'microsoft', 'technical', 'windows', 'service', 'technical support', 'customer']

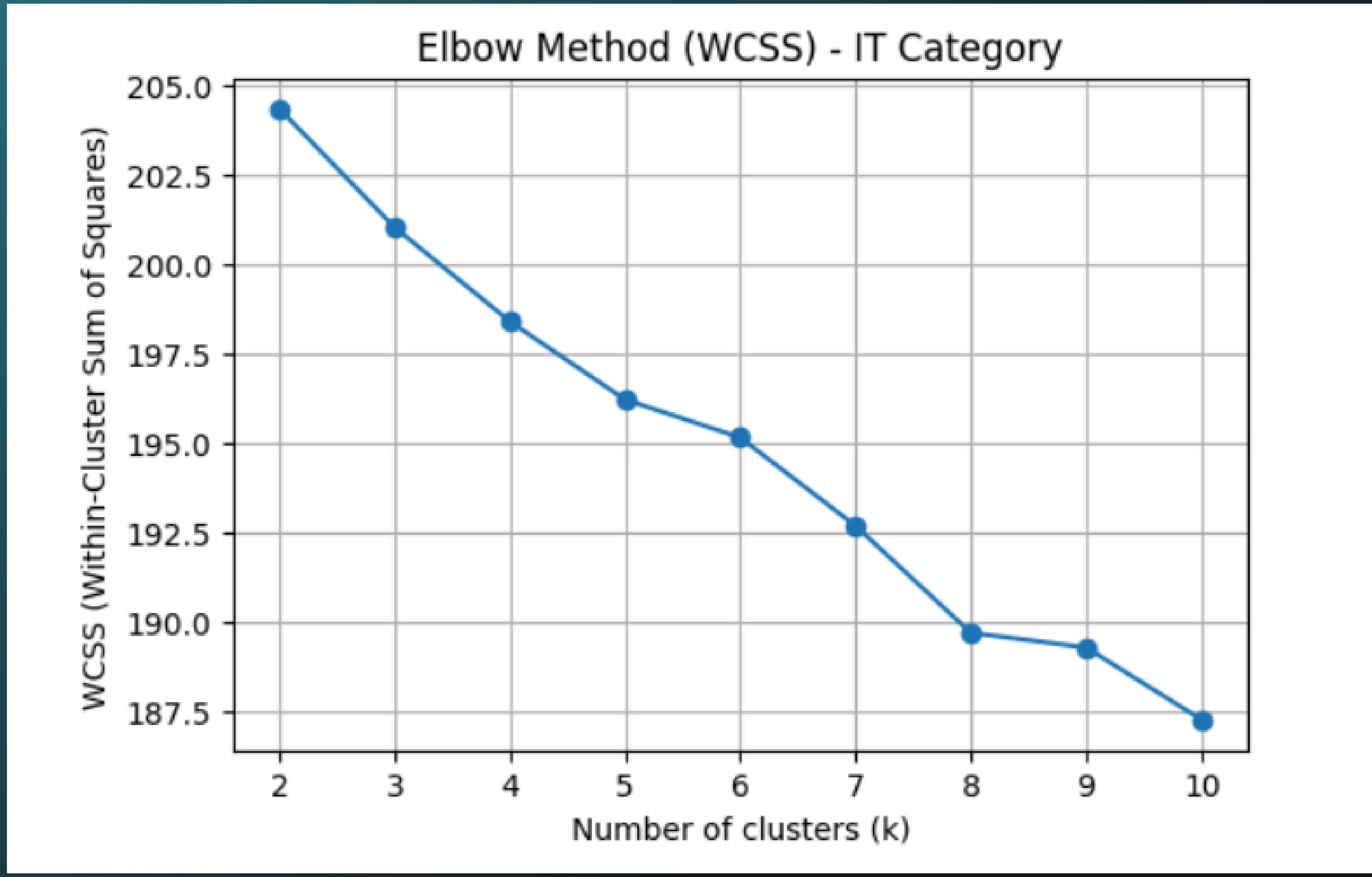
Cluster 4: General IT

Top Keywords

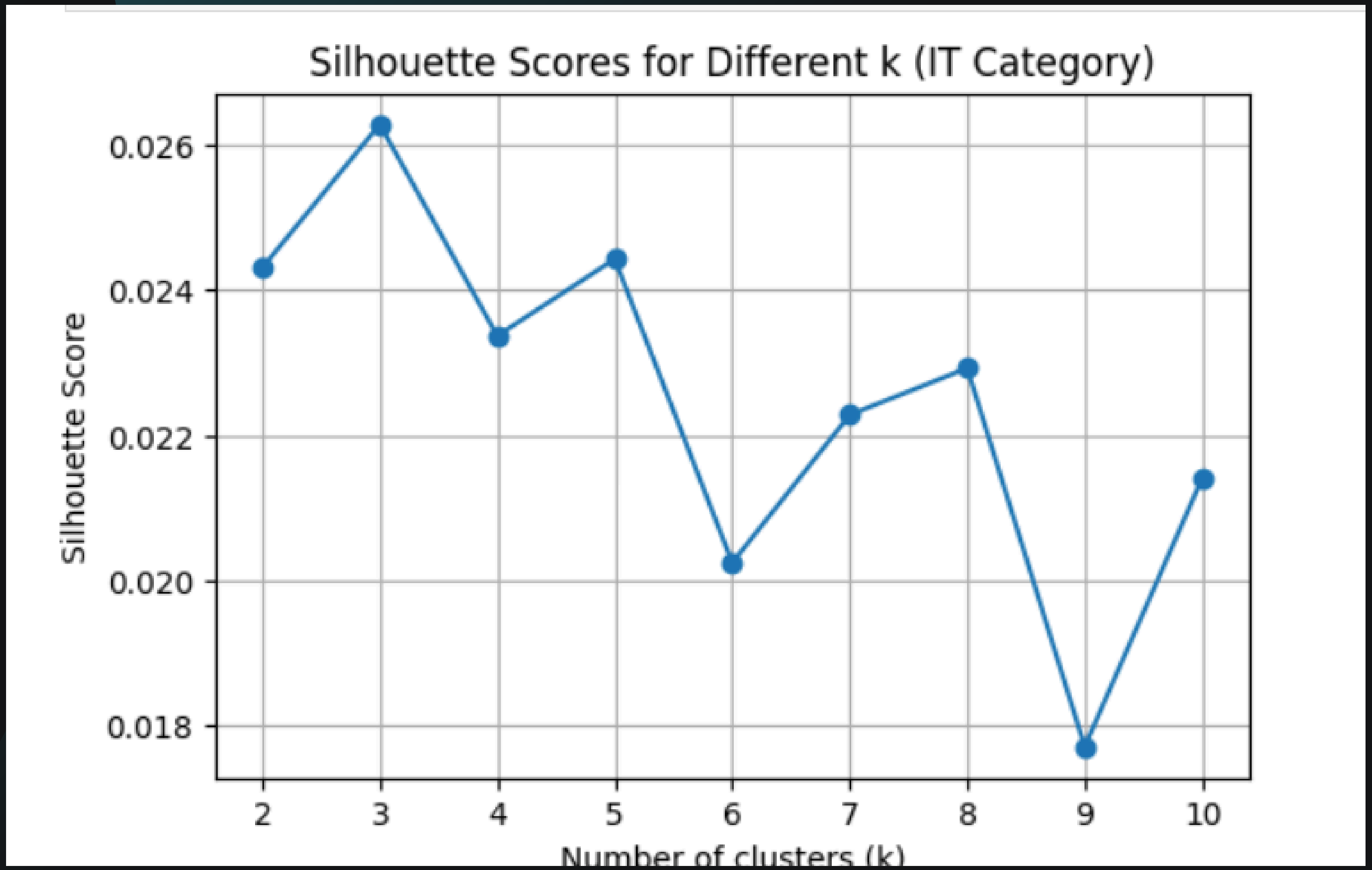
['technology', 'experience', 'information', 'management', 'support', 'work', 'position', 'information technology', 'service', 'systems']

Phase 3: Discovering Hidden IT Specializations

To estimate a suitable number of clusters k for the IT jobs the **Within-Cluster Sum of Squares (WCSS)** is computed for different values of k (from 2 to 10), then The choice of k is further evaluated using the **Silhouette Score.**



Within-Cluster Sum of Squares (WCSS) Chart



Silhouette Score Chart

Phase 4 (Generative AI): Final output Demo

Template 1 :

```
Python
================================================================================
TEMPLATE 1 OUTPUT:
================================================================================
# Career Advice Report for Aspiring Data Analyst

## 1. Career Matching Explanation
Based on your current skills in Python, SQL, Excel, data visualization, and problem-solving, the predicted job title of Data Analyst aligns well with your profile. Data Analysts are responsible for inte

## 2. Missing Skills
While your current skill set is strong, there are a few additional skills that can enhance your candidacy for a Data Analyst position:

- **Statistical Analysis**: Understanding statistical methods is essential for analyzing data accurately.
- **Data Cleaning and Preparation**: Skills in data wrangling and cleaning are vital, as raw data often requires significant preprocessing.
- **Business Acumen**: A strong understanding of business processes and key performance indicators (KPIs) can help in making data-driven decisions.
- **Advanced Data Visualization Tools**: Familiarity with tools like Tableau, Power BI, or similar platforms can greatly enhance your ability to create impactful visualizations.
- **Machine Learning Basics**: While not always required, having a foundational understanding of machine learning concepts can be beneficial.

## 3. Learning Plan
To bridge the gaps in your skills and enhance your profile for a Data Analyst role, consider the following learning plan:

1. **Enroll in a Statistics Course**:
   - Take an online course (e.g., Coursera, edX) that covers basic to intermediate statistical analysis concepts. Focus on descriptive statistics, inferential statistics, and hypothesis testing.

2. **Master Data Cleaning Techniques**:
   - Use resources like Kaggle or DataCamp to learn techniques in data cleaning and preprocessing using Python libraries such as Pandas.
...
## 4. Final Advice
As you pursue a career as a Data Analyst, maintain a strong portfolio showcasing your projects and analyses. Engage in real-world data projects, either through internships, freelancing, or by contributi

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```

Template 2 :

```
Python
================================================================================
TEMPLATE 2 OUTPUT:
================================================================================
Hello, Sara! I'm so glad to see you taking steps towards a career in data analytics. You already have a solid foundation with your skills in Python, SQL, Excel, and data visualization, which are all inc

Data analytics is a fantastic career path for you because it combines technical skills with the ability to tell a story through data. For example, your proficiency in Python and SQL will allow you to ma

To take your career to the next level, here are a few key skills you should focus on developing next:

1. **Statistical Analysis:** Understanding basic statistics will help you interpret data accurately and draw meaningful conclusions. This is crucial in analytics since you'll often need to analyze trend

2. **Data Storytelling:** While you have great technical skills, being able to communicate your findings clearly is equally important. Consider working on your presentation skills and learning how to st

3. **Advanced Excel Skills:** While you're already familiar with Excel, diving deeper into advanced functions, pivot tables, and macros can greatly enhance your data manipulation capabilities. Many orga

Now, let's create a practical learning roadmap for you:

### This Week:
- **Enroll in a Statistics Course:** Look for online platforms like Coursera or edX that offer introductory statistics courses. Aim for at least 2-3 hours of study this week.
- **Practice Data Visualization:** Use a free tool like Tableau Public to create a simple data visualization project. Choose a dataset that interests you and try to tell a story with it.

### This Month:
- **Take an Advanced Excel Course:** Spend a few hours each week on platforms like LinkedIn Learning or Udemy, focusing on advanced Excel features. Set a goal to complete one course by the end of the mo
- **Join a Data Analytics Community:** Engage with others in the field through online forums or local meetups. This is a great way to network, learn from others, and share your progress.
...

You're on a great path, Sara! Keep pushing forward, stay curious, and embrace every learning opportunity that comes your way. I believe in you, and I can't wait to see where your journey in data analyti

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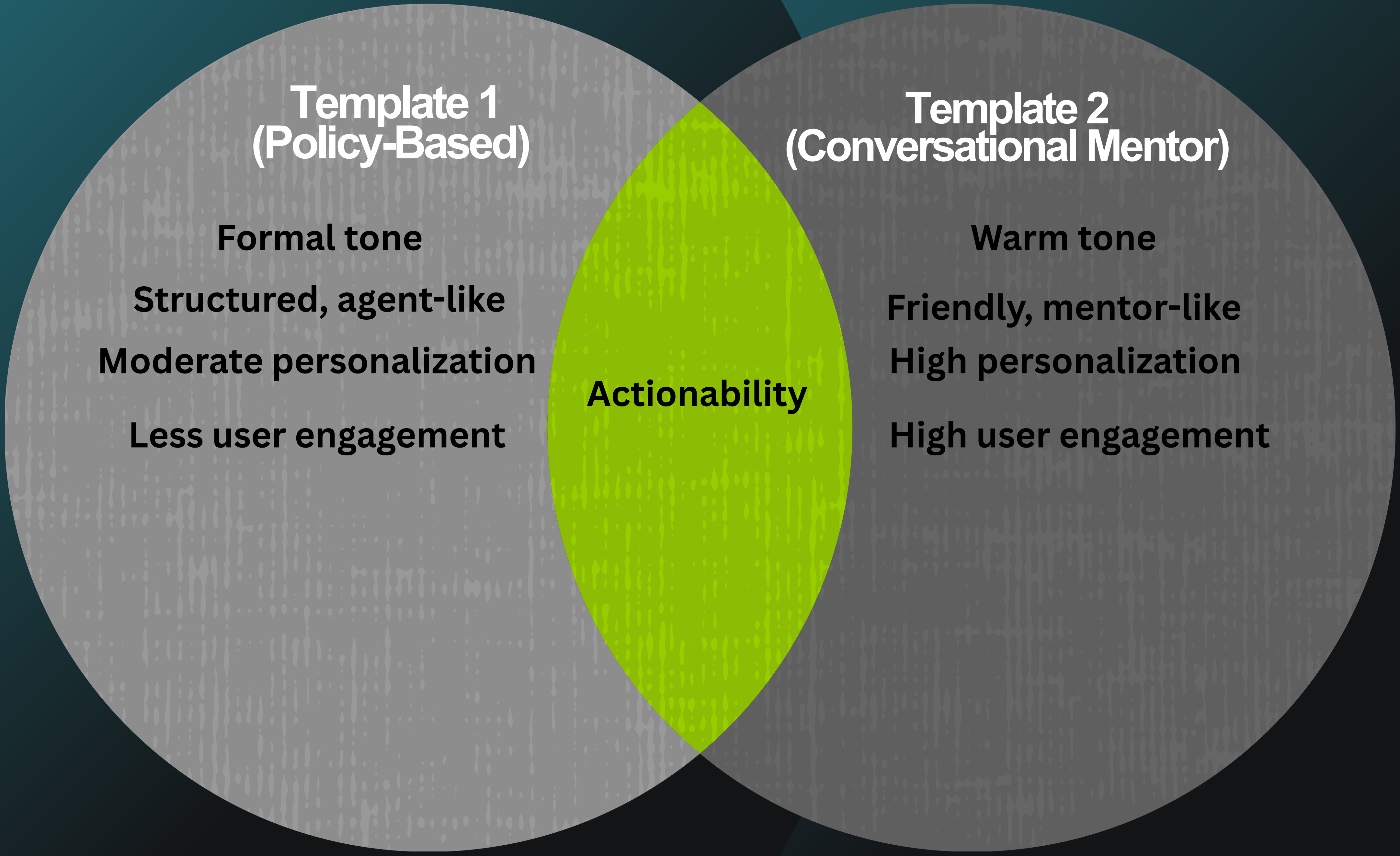
Phase 4 (Generative AI): Comparison between Template 1 and Template 2



Our choice between them is : Template 2 (Conversational Mentor Approach)

Summary

Precision

Determining the exact career field with 95% accuracy

Depth

Discovered hidden career tracks in the unsupervised phase

Empathy

Using Generative AI to turn cold data into warm, actionable advice

Recommendation

1 Deploy in university portals to assist academic advisors

2 Use clustering to update curriculum based on market trends (e.g., increasing demand for Security).

Future Work

1 **CV Parsing**

Upload PDF resumes directly

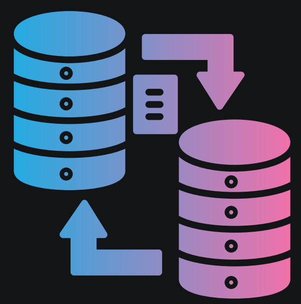
2 **More Industries**

Expand beyond the current 5 categories

3 **Live Data:**

Integrate LinkedIn API for real-time job matching

Lessons Learned

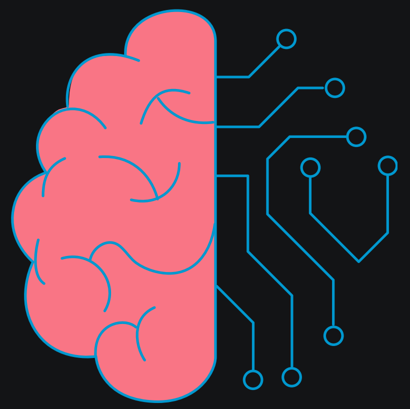


Data Preprocessing is Critical

Raw data was clean, but machines don't understand text.
Success came from intelligently transforming text using TF-IDF and N-grams to capture technical context.

Interpretation

Understanding why a model fails (Error Analysis) is more valuable than raw accuracy



Effective Prompt Design

Generative AI is powerful, but it relies on clear instructions.
We learned that Prompt Engineering (e.g., testing "Recruiter" vs. "Mentor" personas) is important to getting useful, empathetic advice.

Thank You